

BANARAS HINDU UNIVERSITY

B.A. (Hons.) Semester III Examination, 2023-24

Paper No. STA-301: Statistical Inference

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Explain the concept of the sampling distribution of a statistic and standard error. Describe the utility of standard error in hypothesis testing. Derive the expressions for the standard error of the following cases:

- (i) The mean of a random sample of size n obtained from a normal distribution.
- (ii) The difference of the means of two independent random samples of sizes n_1 and n_2 from normal distributions $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ respectively.

Also discuss the case of equal but unknown variance for both populations. [#1]

- (b) Obtain the sampling distribution of the sum of independent Poisson random variates. [#1]

Question 2.

- (a) Discuss the criteria of a good estimator. Let X_1, X_2, X_3, X_4 be a random sample from a $N(\mu, \sigma^2)$ population. Find the variance of $T = \frac{X_1 + 3X_2 + 2X_3 + X_4}{7}$ and $\bar{X} = \frac{X_1 + X_2 + X_3 + X_4}{4}$. Which one is preferred between T and $\bar{X}$ for the estimation of the mean of a normal distribution and why? [#1]

- (b) Show that for a normal distribution, the sample variance $S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is not an unbiased estimate of the population variance σ^2 . Obtain an unbiased estimate of the population variance. [#1]

Question 3.

- (a) Let P_1 and P_2 be the unknown proportions of students wearing glasses in two universities A and B. To compare P_1 and P_2 , samples of sizes n_1 and n_2 are taken from the two populations and the number of students wearing glasses is found to be x_1 and x_2 respectively. Suggest an unbiased estimate of $P_1 - P_2$ and obtain its sampling distribution when n_1 and n_2 are large. Hence, explain how to test the hypothesis $H_0 : P_1 = P_2$. [#1]
- (b) Twenty people were attacked by a disease and only 18 survived. Will you reject the hypothesis that the survival rate, if attacked by this disease, is 85% in favor of the

hypothesis that it is more, at 5% level of significance? (Use large sample test; the tabulated Z value in a one-tailed test is 1.645). [#1]

Question 4. Describe the χ^2 test of significance and state its various applications. What are the assumptions of the χ^2 test of goodness of fit? Discuss its additive property and the limiting form of χ^2 for large degrees of freedom. [#1]

Question 5.

- (a) Explain stating clearly the assumptions involved in the t -test for testing the significance of the difference between two means. Find the value of the t -statistic in a sample of size eight: $-4, -2, -2, 0, 2, 2, 3, 3$, taking the mean of the universe to be zero. How would you proceed further to test at 5% level of significance? (Given $t_{0.05,7} = 2.365$). [#1]
- (b) Discuss the F -distribution along with its applications. [#1]
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Question 6.

- (a) State the Weak Law of Large Numbers (WLLN). Examine whether the WLLN holds or not for the sequence $\{X_n\}$ of independent random variables defined as follows:

$$P(X_n = 2^n) = 2^{-2n-1}, \quad P(X_n = -2^n) = 2^{-2n-1}, \quad P(X_n = 0) = 1 - 2^{-2n}$$

[#1]

- (b) Write the statement of the Central Limit Theorem (CLT) for the i.i.d. case. Also explain its utility. [#1]
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Question 7.

- (a) Define order statistics. What are the cumulative distribution functions of the smallest and largest order statistics? Also, obtain the probability density function of the r -th order statistic. [#1]
- (b) Discuss the applications of non-parametric tests, along with their advantages and disadvantages. Also, explain the Run test. [#1]
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Question 8. Write short notes on any **three** of the following: [#1]

- (a) Errors, Critical Region, Level of Significance, and Power of a test in hypothesis testing.
- (b) χ^2 -test for independence of attributes.
- (c) Kolmogorov-Smirnov (K-S) test.
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(d) Fisher's Z -transformation and its applications.